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Parameter Optimization for Dissimilar Aluminum Alloys Joined Using Friction Stir Additive Manufacturing: A Screening Study

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ABSTRACT

Due to the requirement for better strength-to-weight ratios, the utilisation of aluminum alloys is rapidly expanding. Lightweight components are of utmost importance in most industries, particularly in transportation, aviation, maritime, automotive, and other industries. These lightweight engineering materials are hard to be joined utilising traditional fusion joining techniques, necessitating the development of alternative joining techniques. The novel friction-stir additive manufacturing (FSAM) technique, based on the concept of friction stir welding (FSW), can be used to combine aluminum alloys additively in their solid state. This work examines the effects of several process parameters (tool rotational speed, tool tilt angle, and tool transverse speed) on tensile strength and hardness using a 3-factor L9 Taguchi designed experiment. Three cases were explored, one were AL 6061 was welded to AL 7075, one where AL 7075 was welded to AL 6061, and one where the data were mixed to get an "average" effect representative of large additively manufactured parts. A detailed ANOVA (including both main effects and interactions analyses) provided clear guidance on the optimization of the parameters for several objectives. This work will contribute to the development and wider use of FSAM in both industrial and academic research settings by providing a useful dataset and clear parameter selection guidance. The results of this research indicate that the FSAM methodology could be utilized to fabricate large defect-free structures, which can be a suitable replacement for the traditional Al6061 material used in automotive and aerospace sectors.

1 | Introduction

With the current advancements in science and technology, the necessity for joining materials with improved tensile strength and hardness properties has emerged. Friction stirring is a technique for extruding or plastically deforming the material before it reaches its melting point, where friction between the tool and base material generates heat [1]. Among the numerous advancements of this technique, welding saw its first use and was assigned the name "friction stir welding" (FSW), a relatively recent solid-state joining procedure [2–9]. Defects like porosity, hot cracking, and alloy segregation are not commonly present

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in FSW, and welds degenerate with acceptable surface quality, necessitating no post-weld cleaning [10–16]. This method of welding uses a non-consumable tool, which is harder than the material being welded [17, 18]. This technique is advised for soft materials such as aluminum, titanium, nickel, copper, and other metals, whether ferrous or non-ferrous and of the same kind or different kinds [15, 19–24]. Additive manufacturing has been embraced as a desirable approach to build metallic components of the vehicle and aerospace sectors due to the necessity for sophisticated and particular engineering materials for specific applications [25, 26]. A novel, environmentally friendly, energy-efficient solid-state additive manufacturing (AM) method called friction-stir additive manufacturing (FSAM) offers a lot of potential for synthesizing lightweight materials with excellent structural properties [27–29]. In the FSAM process, metals such as alloys of aluminum and alloys of magnesium are joined in a solid state manner without any melting or recasting. As a result, FSAM is a better option than most currently used beam-based metal additive manufacturing processes since it can manufacture components with few defects, such as porosity, inclusions, and internal cavities [30–32].

Because of its benefits above fusion-based AM approaches, FSAM has attracted focus in solid-state AM in the past few years. It operates on the same principles of friction-stir lap welding (FSLW), where multiple plates are joined due to heat generated as a result of friction and deformation in a plastic manner of the material when a non-consumable rotating tool is moved across the face of a workpiece in a pre-determined direction [33, 34]. Over the past several years, FSAM has demonstrated its ability to join various lightweight metals [26, 28, 32, 35]. Airbus pioneered the idea of employing the friction stir process to join the metals additively in 2006. Nevertheless, due to their early phases of development, there have only been a relatively small number of studies in the years that have followed. The FSW approach includes a critical consideration for tool profile [36]. Palanivel et al. have demonstrated that solid-state additive manufacturing can fabricate high-quality builds from lighter alloy structures by utilising the FSAM technique [25, 37]. They built a four-layered, 5.6 mm-long FSAM structure using WE43 Mg alloy. Due to varying heat levels developed during FSAM, they reported varying microhardness and microstructure through the build depth. Yueqing et al. utilised AL7075-alloy to construct a 42 mm construct length FSAM structure [38]. They identified a finely-grained microstructure in the final part and recorded a peak hardness value of 115 HV in the upper layer of the built structure. Zhao et al. produced a five-layer FSAM structure made of 2195-T8 aluminum-lithium alloy, and the best results were achieved at process parameters of 800 rpm tool rotation speed (TRS) and a tool traverse speed (TTS) of 100 mm/min [39]. However, the highest UTS found had only been 56% of the base metal due to the annealing effect. Changshu et al. have built a 42 mm construct by FSAM using a 7N01-T4 alloy [40]. They observed that the existence of kiss bonds at the interface resulted in a reduction in UTS and microhardness. Multilayered builds of Al 5083 [41], Al 6061-T6 [42], copper [43], Al 2060 [44] Al-Si-Mg alloy [45] were also reported by various authors who had utilised FSAM technique.

However, research on FSAM using dissimilar Al alloys is meager so far. The only significant work in the literature so far was done by Venkit and Selvaraj [34, 46]. In their two works on this topic, they developed the manufacturing technique for mixing layers of Al 6061 and Al 7075, two extremely important metals for the aerospace sectors. In these studies, laminate composite materials were developed and tested for their properties and performance. However, a major area that was not covered in these previous studies by Venkit and Selvaraj or in the previously mentioned work on FSAM was parameter exploration and optimization. The present work picks up here, developing a designed experimental approach for testing the effects of the process parameters and helping to guide better exploration of this process in the future. Three factors (tool rotational speed, tool tilt angle, and transverse speed) were studied using a 3-factor L9 Taguchi experimental design (three levels per factor). The results were examined using ANOVA to determine the contribution of each factor on the output (ultimate tensile strength and Vickers hardness).

2 | Materials and Methods

Aluminum alloy plates of 250 mm × 150 mm × 3 mm of Al 7075-T6 and Al 6061-T6 are utilized as the build material. The friction stir AM (FSAM) process used is shown in Figure 1a. Here the principle of FSLW is utilized to bond alternate metal

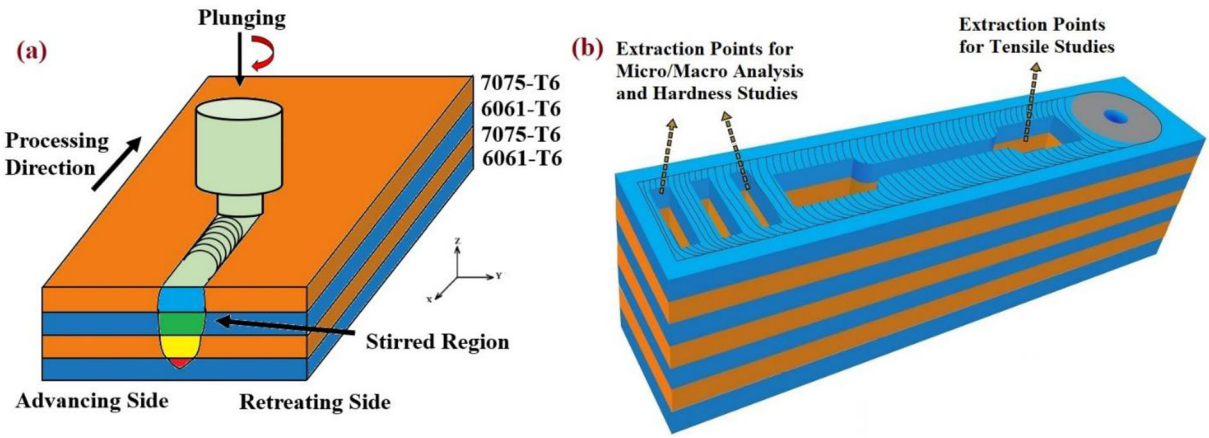

FIGURE 1 | (a) Illustration of FSAM technique and (b) Extraction points of samples. Adapted from [34], updated and reproduced under a CC-BY license.

plates one over the other by piling a new plate above the formerly lap-welded plates. After joining two plates, the top surface of the combined material is milled flat to align the next plate on top of it. These procedures were continued until the required height of build was achieved. Two distinct tools were utilised to avoid unintended deposition of materials from one plate to the other during the FSAM procedure since the alloying elements in the two plates were different. An H13 tool steel with a conical threaded pin with 24 mm shoulder diameter, 4 mm probe height, and probe diameter tapering from 8 mm to 6 mm was used in this procedure. A continuous vertical force of 20 kN was applied to the structure throughout the FSAM procedure to ensure accurate material compaction and to eliminate porosity as much as possible.

Initially, before the FSAM process, parameter optimisation of the tool's tilt angle (TTA), TRS (RPM), and TTS was performed independently for Al 6061 and Al 7075 when these plates were positioned on top. After multiple trial welds, a reasonable operating range of various tool parameters was identified and is shown in Table 1. The selection of lower TRS (less than 1100 rpm) resulted in the formation of wormholes, and higher TTS (greater than 40 mm/min) of tool resulted in tunneling defects. After fabricating the FSAM component, tensile samples were extracted in the longitudinal direction, and samples for checking hardness and microstructure were extracted from the weld area as shown in Figure 1b. Samples were cut from the welded block using a wire EDM machine. The sample geometry for each are shown in Figure 2. The tensile samples were done according to ASTM E8/E8M-22 (displacement speed of 1 mm/min) and the hardness samples according to ASTM E384. The hardness tests were done at intervals of 0.25 mm along the length of the samples.

TABLE 1 | Experimental factors and levels for a three-factor L9 Taguchi experiments design.

Factor	Level 1	Level 2	Level 3
TRS: Tool rotation speed (RPM)	1100	1150	1200
TTA: Tool tilt angle (°)	1.0	1.5	2.0
TRA: Tool transverse speed (mm/min)	30	35	40
3-factor L9 Taguchi Design	**Factor A**	**Factor B**	**Factor C**
Condition 1	1	1	1
Condition 2	1	2	2
Condition 3	1	3	3
Condition 4	2	1	2
Condition 5	2	2	3
Condition 6	2	3	1
Condition 7	3	1	3
Condition 8	3	2	1
Condition 9	3	3	2

Based on the reasonable ranges of process parameters which resulted in sound parts (Table 1) established before, a 3-factor L9 Taguchi experimental design was constructed to explore the effects of each parameter and provide guidance toward their optimization. For each combination of factors studied, a tensile test and three Vickers hardness tests were completed for two core cases (6061 over 7075 and 7075 over 6061). Samples from the raw materials (two runs each) were also used as baseline comparisons. Given this setup, a total of 22 tensile tests and 66 Vickers hardness tests were completed in this study.

3 | Results

The first tests completed were those on the base materials. Table 2 shows the results. Each of the base materials were tensile tested twice, while the hardness was tested three times for each. After this, the samples for the case where Al 6061 was welded onto Al 7075 and the case where Al 7075 was welded into Al 6061 were tested. The results are shown in Tables 3 and 4, respectively. The data is also tabulated in Figure 3.

4 | Statistical Analysis

After completion of the tests, an analysis of variance (ANOVA) was completed (Minitab v20.3) in order to determine the main effects and interactions between the factors in the Taguchi experiment. This was done four times, one for each of the layer orders for tensile strength and hardness. The p-values are given in Table 5, supported by the main effects plots in Figure 4 and interaction plots in Figure 5. In order to better explore the "average" case where there is a mix of the material order (as would be seen

TABLE 2 | Experimental responses (ultimate tensile strength (UTS) and Vickers hardness (VH)) for the base materials.

Material	UTS (MPa) (Mean)	UTS (MPa) (Stdev)	VH (Mean)	VH (Stdev)
Al 6061	321	15.56	122.1	0.20
Al 7075	572	14.85	194.1	1.33

Note: Tensile data was from two runs, while hardness data was the average of three automated runs.

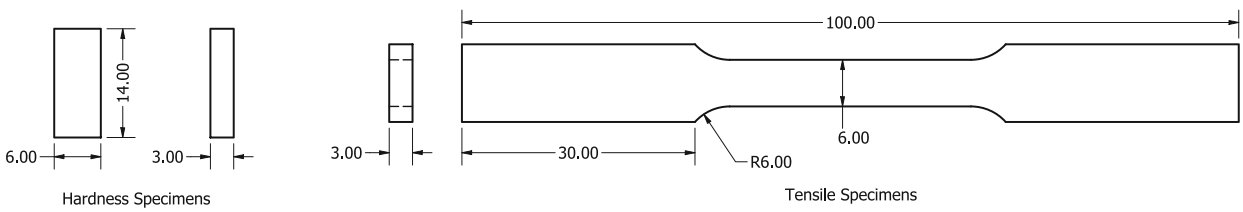

FIGURE 2 | Specimen dimensions (all dimensions in mm).

TABLE 3 | Experimental responses (ultimate tensile strength (UTS) and Vickers hardness (VH)) for Al 6061 over Al 7075 sample plates.

Run	TRS (RPM)	TTA (°)	TTS (mm/min)	UTS (MPa)	VH (Mean)
1	1100	1.0	30	287	104.5
2	1100	1.5	35	295	116.9
3	1100	2.0	40	330	120.0
4	1150	1.5	30	318	87.1
5	1150	2.0	35	326	104.9
6	1150	1.0	40	308	69.5
7	1200	2.0	30	320	109.8
8	1200	1.0	35	290	87.5
9	1200	1.5	40	298	98.4

Note: Tensile data was from one run, while hardness data was the average of three automated runs.

TABLE 4 | Experimental responses (ultimate tensile strength (UTS) and Vickers hardness (VH)) for Al 7075 over Al 6061 sample plates.

Run	TRS (RPM)	TTA (°)	TTS (mm/min)	UTS (MPa)	VH (Mean)
1	1100	1.0	30	291	112.2
2	1100	1.5	35	297	120.3
3	1100	2.0	40	325	135.2
4	1150	1.5	30	307	115.1
5	1150	2.0	35	320	125.3
6	1150	1.0	40	322	97.6
7	1200	2.0	30	336	127.6
8	1200	1.0	35	326	105.0
9	1200	1.5	40	330	115.0

Note: Tensile data was from one run, while hardness data was the average of three automated runs.

TABLE 5 | Calculated p for each case in ANOVA.

Source	UTS (6061/7075)	UTS (7075/6061)	VH (6061/7075)	VH (7075/6061)
A: RPM	0.224	0.036	0.009	0.094
B: TTA	0.077	0.084	0.012	0.017
C: TTS	0.520	0.136	0.222	0.790

in realistic AM), the analysis was repeated with the two datasets (Al 6061 welded to Al 7075 and vice-versa) combined. The results of this are shown in Table 6 and Figure 6.

The purpose of this study was to explore the effects of the different parameters using a 3-factor L9 Taguchi designed experiment. A total of 22 tensile tests and 66 hardness tests were collected. The data analysis in the previous section showed that some of the parameters had a statistically significant effect on the tensile strength and hardness of the materials. Which factors did depends on the choice of level of significance. At a typical $\alpha = 0.05$, none of the parameters had an effect on the UTS for the 6061/7075 case, but the RPM did for the 7075/6061 case. In contrast, TTA had a significant impact on both cases for the hardness, while RPM also had an effect on the 6061/7075 case. When the datasets were mixed (as would be commonly seen with a large AM part), the TTA was also found to have a significant impact on both UTS and hardness. However, since a Taguchi design contains a relatively small number of datapoints and the experiment would contain a lot of noise, a case could be made for a higher level of significance. Levels of $\alpha = 0.1$ are also common in manufacturing research for this reason. The level of significance used would be a decision for the stakeholders. A higher α would increase the power of the experiment but also increase the probability of making a false positive (Type I error). In this case, the consequences of a false positive would be a slightly less efficient manufacturing process, while the consequences of making

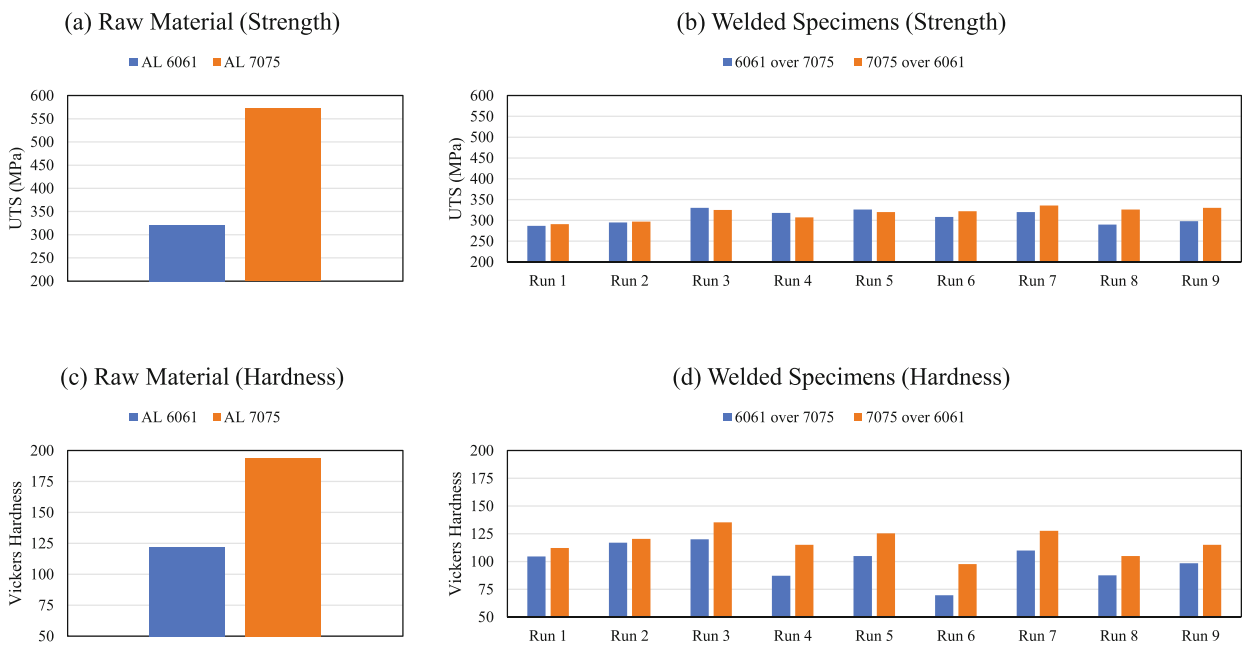

FIGURE 3 | Plotted results compared with base materials. Note that the plot scales are equal to show the softening effects of FSAM.

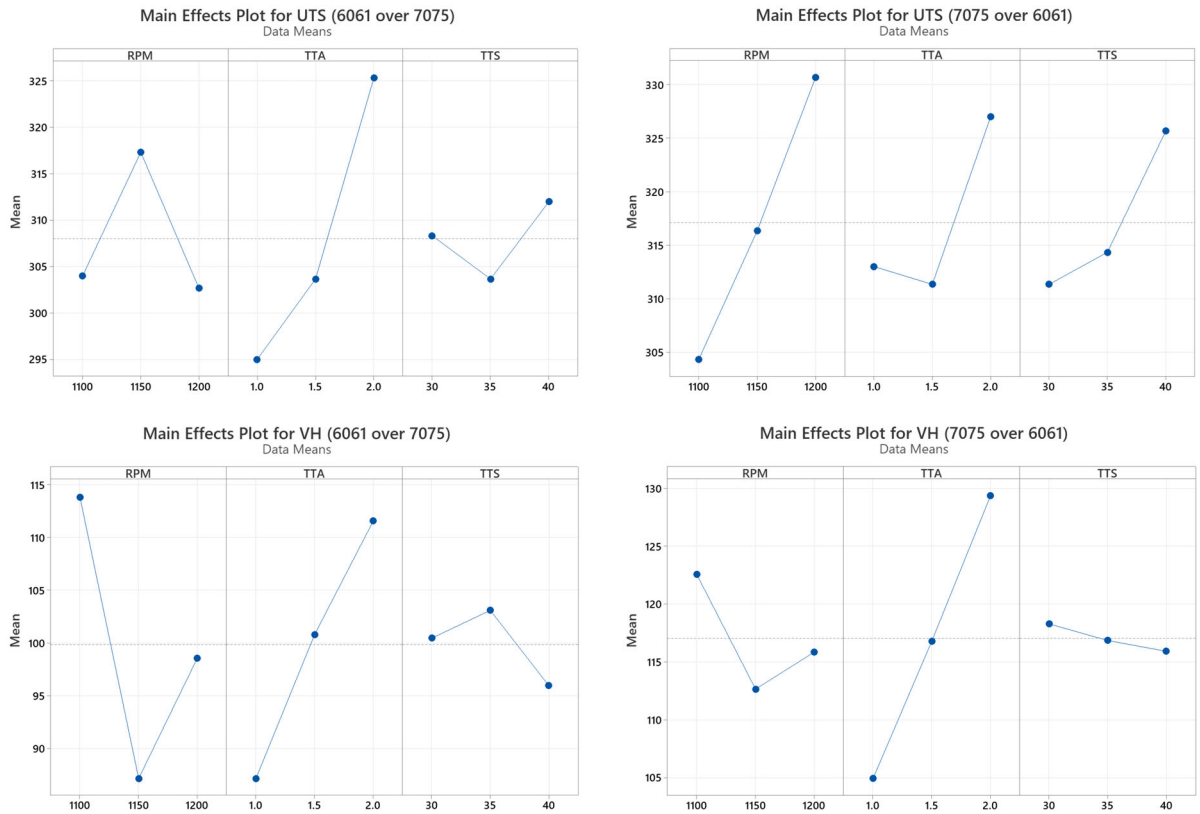

FIGURE 4 | Main effects plot for each case.

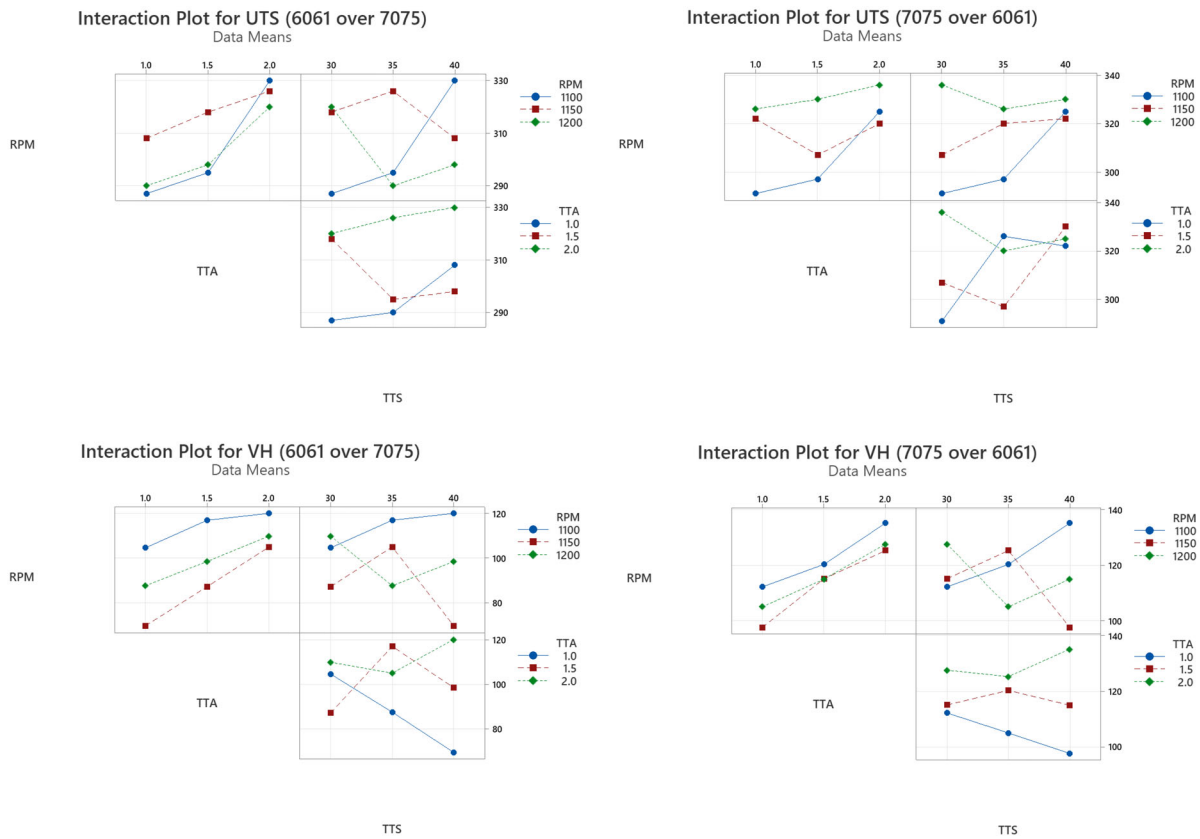

FIGURE 5 | Interactions plot for each case.

a Type II error (false negative) are far more dire. Therefore, the risk of using a larger level of significance is probably a wide idea for the stakeholders deciding on parameters.

5 | Formation Characteristics and Microstructure

This screening study focused on the statistical analysis and practical application of this method, but a discussion of the microstructure and formation characteristics would benefit the conclusions. Previous work [33, 34, 46] by members of the author team examined the microstructure in detail, using both microscopy and SEM images. Relevant images are shown in Figures 7, 8, and 9. Figure 7 clearly shows the nature of the interfacial bond between the layers due to the differing metals used in each layer in the structure. The final FSAM bulk material exhibits a number of microstructural zones, including heat-affected zones (HAZ), thermo-mechanically affected zones (TMAZ), and stir zones (SZ). All layers of the SZ (Figure 8) shows an onion ring pattern created due to upward material movement and extrusion. The contrast which is visible inside the ring pattern may be attributed chiefly to the presence of Al 6061 and Al 7075 in the SZ. Figure 8 clearly shows the presence of artifacts such as hooks and kiss bonds in different layers of the FSAM component. Hooks, which is a natural phenomenon visible on RA and AS, are unbonded connections created by the deformation of surfaces. In RS, the hook spreads to the SZ, but in the AS, it reaches toward the HAZ and TMAZ. The primary reason for the formation of kiss bonds is attributed to the insufficient flow of material owing to the incomplete stirring by the tool's pin and also due to the creation of oxides.

It is clear from microstructure images (Figure 9a–c) that none of the regions had any mechanical flaws, such as voids or micropores. None of the layers showed any volumetric flaws that would compromise their mechanical properties. From microscopy images, a uniform blending of different materials can be seen as well. The SZ micro images show that fine grains are equiaxed due to dynamic recrystallization. When compared to SZ, the grain sizes of TMAZ, which is mostly impacted by the shoulder, are noticeably larger. A transition area is observed that is characterised by the development of banded formations with mainly coarser grains. It has been observed that grain size

TABLE 6 | Calculated p for the mixed data case in ANOVA.

Source	UTS (Mixed)	VH (Mixed)
A: RPM	0.178	0.074
B: TTA	0.020	0.013
C: TTS	0.349	0.990

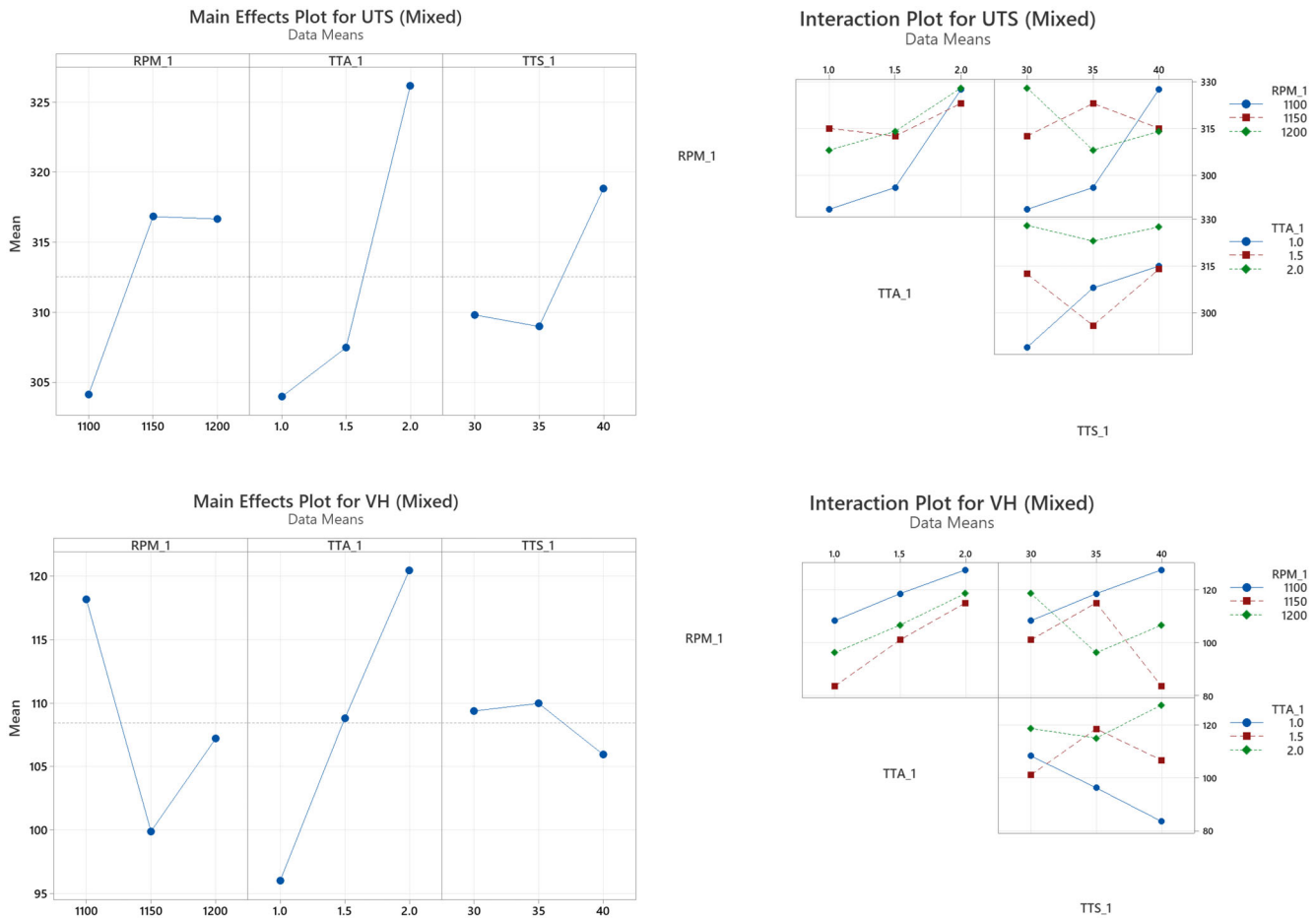

FIGURE 6 | Main effects and interactions plots for the mixed case.

increases from the top to the bottom of the structure when moving throughout the path of the structure thickness. FE-SEM analysis was done to investigate the dissimilarities in the secondary phase distribution on the final FSAM structure. During the FSAM technique, the structure is severely plastically deformed and subjected to high temperatures. During this period, the coarser particles that serve as reinforcing elements are vulnerable to melting, breaking apart, and the formation of precipitates. Minute precipitated particles are formed as a consequence of these processes and are dispersed throughout the SZ in a random fashion, as seen in Figure 9d–f. EDX analysis revealed that $MgZn_2$ and AlZnMgCu particles precipitated from Al 7075 and $AlMg_2Si$ particles precipitated from Al 6061.

6 | Parameter Optimization

When reviewing the main effects plots (Figures 4 and 6), there is clear information about the appropriate optimization settings for each parameter. This holds true for both the layer-order cases and the mixed case. Therefore, a decision-making table based on the optimized parameters can be created. This table, with the six possible objectives outlined, is shown in Table 7. It should be noted that the interactions analysis (Figures 5 and 6) show that there is very little to no interaction between the RPM and TTA, suggesting that they are independent parameters that do not have a combined effect. However, this is not the case for RPM with TTS and TTA and TTS; in these cases, there is a strong interaction. This must be considered when making decisions about parameters, as these are not independent parameters and have to be considered together.

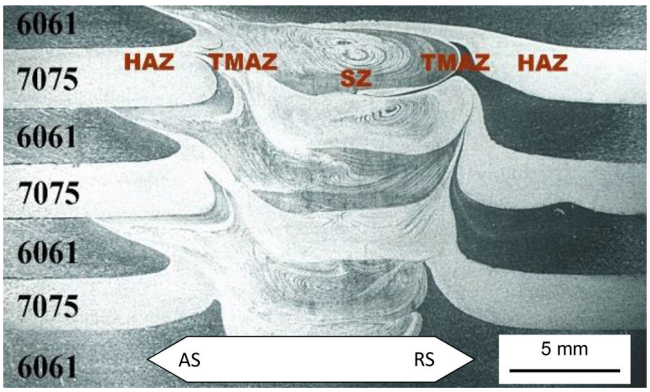

FIGURE 7 | FSAM build cross-section view. Adapted from [34], reproduced under a CC-BY license.

7 | Conclusions and Closing Remarks

This communication explored the effects of the input parameters on the welding of dissimilar aluminum alloys for friction stir additive manufacturing. This work carried on the previous process development and microstructural analyses carried out by Venkit and Selvaraj [34, 46], looking at the best way to optimize the parameters. The parameters of interest were the rotational speed (RPM), the tool tilt angle (TTA), and tool transverse speed (TTS). These were studied using a 3-factor L9 Taguchi designed experiment, both to see which (if any) of the parameters showed a statistically significant impact and to see the optimal selection of parameters for several different objectives related to tensile strength and hardness. After looking at two major cases where one alloy was welded onto the other, the data were mixed to get information on the "average" expected behavior using additive manufacturing of multiple layers. The results gave clear guidance on parameter selection, and parameter interactions for the two materials selected, as well as the mixed data case (which approximates a realistic AM process). These results serve as a useful screening study to provide guidance on how to use the FSAM process with aluminum alloys in the future, relevant for both industrial and academic research applications. The results of this

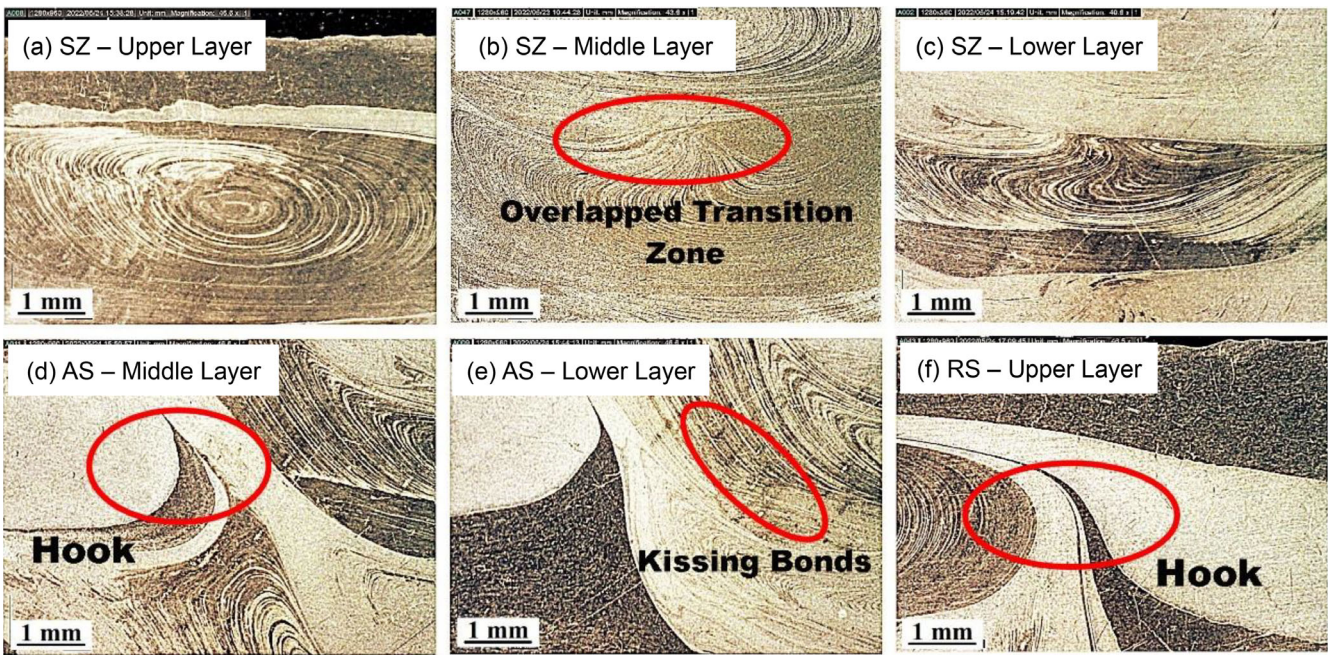

FIGURE 8 | FSAM Formation morphology in various locations. Adapted from [34], reproduced under a CC-BY license.

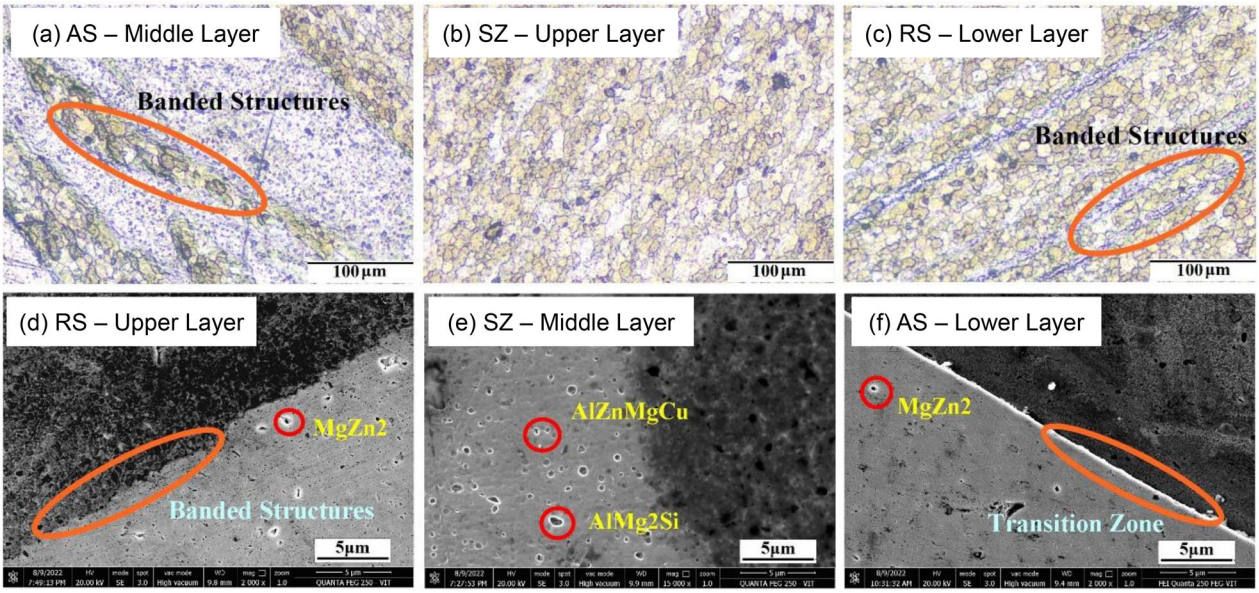

FIGURE 9 | (a–c) Microstructure images in various locations and (d–e) FE-SEM images of precipitate distribution in various locations. Adapted from [34], reproduced under a CC-BY license.

TABLE 7 | Parameter optimization recommendations.

Objective	RPM	TTA	TTS
Maximize UTS with Al 6061 welded to Al 7075	Moderate	High	High
Maximize UTS with Al 7075 welded to Al 6061	High	High	High
Maximize VH with Al 6061 welded to Al 7075	Low	High	Moderate
Maximize VH with Al 7075 welded to Al 6061	Low	High	Low
Maximize UTS Mixed Case	Moderate	High	High
Maximize VH Mixed Case	Low	High	Moderate

study are expected to further promote the use of FSAM and make it a more trusted and useful process for end-user manufacturing of large parts, particularly in the aerospace industry. Future work in this area should focus on the use of different parameters, a larger dataset, hardness distribution modeling throughout the part, and on concrete mapping to the design of FSAM parts.

Author Contributions

Senthil Kumaran Selvaraj: conceptualization; methodology; resources; supervision; project administration. **Akash L. Manoj:** data curation; writing – original draft; visualization; formal analysis. **Alan B. Mathew:** writing – original draft; visualization; formal analysis; data curation. **Akash V. Govind:** writing – original draft; visualization; formal analysis; data curation. **Govindaswamy Sundaramali:** writing – original draft; visualization; formal analysis; data curation. **Utkarsh Chadha:** investigation; writing – original draft; writing – review and editing; supervision; formal analysis. **Bhaskar Vajipeyajula:** validation; formal analysis; project administration; writing – original draft; writing – review and editing. **Albert E. Patterson:** validation; writing – review and editing; writing – original draft; formal analysis; project administration.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All collected data is presented in table form in the manuscript.

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